

## **Assessment 1**

### **Testing on live application :-**

**Theme: Online Learning Platform**

#### **Section A: Scenario Based Conceptual Questions**

**1. An online learning platform releases new courses every month. Identify the most suitable software development life cycle model and justify your choice.**

Ans :- most Suitable SDLC Model is agile (scrum) model for online learning platform.

Justification:

- New courses are released every month, requiring frequent updates.
  - Agile supports iterative development with short sprints.
  - Customer feedback can be incorporated quickly.
  - Testing is performed continuously during development.
  - Faster delivery of new features and bug fixes.
- Hence, agile (scrum) model is best choice.

**2. For a course purchase feature, define testing objectives from a quality perspective.**

Ans :- Testing Objectives for Course Purchase Feature :-

- ❖ Verify users can successfully purchase courses.
- ❖ Ensure secure payment processing.
- ❖ Validate different payment methods (UPI, Card, Net Banking, Wallet).
- ❖ Verify successful order confirmation and invoice generation.
- ❖ Ensure failed payments do not create duplicate orders.
- ❖ Check proper error messages for invalid transactions.
- ❖ Ensure purchase history is updated correctly.
- ❖ Verify data integrity and security.

### **3. Explain how non-functional testing contributes to better learner experience.**

Ans :- Contribution of Non-Functional Testing to Learner Experience  
Non-functional testing improves the platform by ensuring:

- Performance Testing: Pages load quickly even with many users.
- Load Testing: Platform works smoothly during peak enrollments.
- Security Testing: Protects learner data and payment information.
- Usability Testing: Easy navigation for students.
- Compatibility Testing: Works across browsers and mobile devices.
- Reliability Testing: Prevents crashes during video lectures or quizzes.

This results in a faster, safer, and more user-friendly learning experience.

### **4. During sprint execution, how should testers ensure acceptance criteria are testable?**

Ans :-

Ensuring Acceptance Criteria are Testable During Sprint Execution  
Testers should:

- Review user stories during sprint planning.
- Ensure acceptance criteria are clear, measurable, and unambiguous.
- Ask developers/Product Owner for clarification if requirements are unclear.
- Convert each acceptance criterion into test scenarios.
- Prepare test cases before development is completed.
- Participate in daily stand-up meetings to resolve issues early.

This helps reduce misunderstandings and ensures the feature meets business requirements.

**5. A payment failure occurs after deployment. Assign severity and priority with justification.**

Ans :-

**Severity** : Critical

**Priority** : High

Justification:

- Users cannot complete purchases.
- Direct impact on business revenue.
- Affects customer trust and user experience.
- Requires immediate production fix.

Since it impacts many users and affects business revenue, it should be fixed immediately. Therefore, the severity is Critical and the priority is High.

**Section B: Practical Manual Testing Tasks**

**1. STLC (Student Login): Document the full lifecycle including Requirement Analysis, Planning, Test Design, Execution, and Closure. Define specific Entry and Exit criteria for the Student Login module.**

Ans :- STLC (Student Login Module)

**Phase 1: Requirement Analysis**

- Study login requirements.
- Identify username/password validation.
- Identify functional and security requirements.

**Phase 2: Test Planning**

- Define testing scope.
- Estimate effort.
- Allocate resources.
- Select testing techniques.

### **Phase 3: Test Design**

- Prepare test scenarios.
- Write test cases.
- Prepare test data.
- Review test cases

### **Phase 4: Test Execution**

- Execute test cases.
- Record Pass/Fail.
- Log defects.
- Retest fixed defects.

### **Phase 5: Test Closure**

- Prepare Test Summary Report.
- Record lessons learned.
- Archive test artifacts

**2. Test Design (Course Rating 1-5): Apply Equivalence Partitioning and Boundary Value Analysis to the 1–5 star rating input. List the specific Valid and Invalid numeric test data required to verify the input limits.**

Ans :- Test Design (Course Rating: 1–5 Stars)

#### **Equivalence Partitioning (EP)**

Valid Partition

1

2

3

4

5

Invalid Partition

0

6

-1

7

1.5

5.5

Blank

Null

### **Boundary Value Analysis (BVA)**

| Test Data | Expected Result |
|-----------|-----------------|
|-----------|-----------------|

|   |         |
|---|---------|
| 0 | Invalid |
|---|---------|

|   |       |
|---|-------|
| 1 | Valid |
|---|-------|

|   |       |
|---|-------|
| 2 | Valid |
|---|-------|

|   |       |
|---|-------|
| 4 | Valid |
|---|-------|

|   |       |
|---|-------|
| 5 | Valid |
|---|-------|

|   |         |
|---|---------|
| 6 | Invalid |
|---|---------|

**3. Test Cases (Course Experience): Write one functional test case for each: Course Search & Browse, Course Learning Experience, and Assessments & Quizzes. Each case must include Steps, Expected Result, and Pass/Fail Status.**

Ans :-

**Test Case 1 :** Course Search & Browse

**Test Case ID:** TC\_001

**Module:** Course Search & Browse

**Objective:** Verify that the user can search for a course successfully.

Steps:

1. Open the Online Learning Platform.
2. Enter "Software Testing" in the search box.
3. Click the Search button.

**Expected Result:** The matching course should be displayed in the search results.

**Actual Result:** As expected

**Status:** Pass

## **Test Case 2 : Course Learning Experience**

**Test Case ID: TC\_002**

**Module:** Course Learning Experience

**Objective:** Verify that the user can access and watch the course videos.

**Steps:**

1. Login with valid credentials.
2. Open "My Courses."
3. Select an enrolled course.
4. Play the course video.

**Expected Result:** The video should play successfully without any interruption.

**Actual Result:** As expected

**Status:** Pass

## **Test Case 3 :- Assessments & Quizzes**

**Test Case ID: TC\_003**

**Module:** Assessments & Quizze

**Objective:** Verify that the user can submit a quiz successfully.

**Steps:**

- 1) Login to the application.
- 2) Open the enrolled course.
- 3) Start the quiz.
- 4) Answer all questions.
- 5) Click the Submit button.

**Expected Result:** The quiz should be submitted successfully and the score should be displayed.

**Actual Result:** As expected

**Status:** Pass

**4.Defect Log: Document 3 unique defects found in the modules above. Each report must include: Defect Title, Severity, Steps to Reproduce, Actual vs. Expected Result, and Environment details.**

**Ans :-**

**Defect Log :-** Online Learning Platform

**Defect ID:** DEF-001

**Defect Title:** Student Login accepts invalid password

**Module:** Student Login

**Severity** High

**Priority** High

Steps to Reproduce:-

1. Open the Login page.
2. Enter a valid username/email.
3. Enter an incorrect password.
4. Click Login.

**Actual Result:-** User is logged into the account despite entering an incorrect.

**Expected Result :-**The system should reject the login attempt and display an error message such as "Invalid username or password."

**Environment :-** Windows 11, Google Chrome v126, Test Environment (QA), Build v1.0

**Defect ID** DEF-002

**Defect Title :-** Course rating accepts values greater than 5

**Module :-** Course Rating

**Severity** Medium

**Priority** Medium

Steps to Reproduce:-

1. Login to the application.
2. Open any completed course.
3. Enter a rating value of 6.
4. Click Submit Rating.

**Actual Result :-**The system accepts the rating value of 6 and saves it successfully.

**Expected Result :-** The system should only accept ratings between 1 and 5 and display a validation message for invalid values.

**Environment :-** Windows 11, Google Chrome v126, Test Environment (QA), Build v1.0

**Defect ID** DEF-003

**Defect Title:-** Quiz submission fails after clicking Submit

**Module** Assessments & Quizzes

**Severity** Critical

**Priority** High

Steps to Reproduce :-

1. Login to the application.
2. Open an enrolled course.
3. Start a quiz.
4. Answer all questions.
5. Click Submit.

**Actual Result:-** The quiz is not submitted, and no score or confirmation message is displayed. User responses are lost after refreshing the page.

**Expected Result** The quiz should be submitted successfully, the score should be displayed, and responses should be saved in the system.

**Environment :-** Windows 11, Google Chrome v126, Test Environment (QA), Build v1.0

## **Section C: -Mini Capstone Project – End-to-End Testing Case**

### **Study Context**

**1. Test Planning Document: Prepare a Test Plan covering scope, objectives, testing types, environment, risks, assumptions, and entry and exit criteria.**

**Ans.**

### **Test Plan**

**Project Name :-** Online Learning Platform

**Objectives :-**

- 
- Verify that all major functionalities work as per requirements.



- Ensure users can successfully register, log in, purchase courses, and access learning content.
- Identify defects before production release.
- Validate system performance, security, and usability.

## **Scope**

### **In Scope**

- Student Login
- Course Search & Selection
- Course Purchase
- Payment Gateway
- Course Access
- Course Rating

### **Out of Scope**

- Admin Module
- Instructor Dashboard
- Third-party Payment Gateway internal processing

## **Testing Types**

- Functional Testing
- Integration Testing
- System Testing
- Regression Testing
- Smoke Testing
- User Acceptance Testing (UAT)
- Performance Testing
- Security Testing
- Usability Testing

## **Test Environmen**

- Operating System: MacOs / Android 14
- Browser: Google Chrome
- Internet: Stable Wi-Fi Connection
- Database: Test Databases

## **Risk**

- Delay in receiving stable builds.
- Payment gateway downtime.
- Requirement changes during testing.
- Limited testing time before release.

## **Assumptions**

- Requirements are finalized.
- Test environment is stable.
- Test data is available.
- Payment gateway sandbox is operational.

## **Entry Criteria**

- Requirements approved.
- Test Plan approved.
- Test environment ready.
- Stable application build available.
- Test cases reviewed.

## **Exit Criteria**

- All planned test cases executed.
- No Critical or High Severity defects remain open.
- Test Summary Report completed.
- Product Owner approval received.

**2. Requirements Traceability Matrix: Create a Requirements Traceability Matrix mapping requirements to test scenarios, test cases, and defects.**

| R.id. | Requirement     | Test Scenario                         | Test Case ID | DefectID |
|-------|-----------------|---------------------------------------|--------------|----------|
| R001  | Student Login.  | Verify successful login               | TC001        | D001     |
| R002  | Course Search   | Verify course search                  | TC002.       | D002     |
| R003  | Course Purchase | Verify successful payment             | TC003        | D003     |
| R004. | Course Access   | Verify purchased course is accessible | TC004        | D004     |

### **3. End-to-End Test Scenario: Write one detailed End-to-End test scenario**

**covering:**

**Login → Course Selection → Payment → Course Access**

**Ans.**

**Test Scenario ID :-** TS01

**Scenario Name :-**

Verify that a student can successfully purchase and access a course.

**Pre-condition :-**

- Student account is registered.
- Internet connection is available.

**Test Steps :-**

1. Open the Online Learning Platform.
2. Login using valid Email and Password.
3. Search for the "Software Testing" course.
4. Open the course details page.
5. Click Buy Now.
6. Select UPI as the payment method.
7. Complete the payment successfully.
8. Open My Courses.
9. Select the purchased courses.

**Expected Result :-**

- Login is successful.
- Course is displayed.
- Payment is completed successfully.
- Purchased course appears in My Courses.
- Student can watch the course without any issue.

**Actual Result:-** All steps are executed successfully.

**Status :-** Pass

**4. Test Summary Report: Prepare a Test Summary Report including execution status, defects identified, and release recommendation.**

**Ans :-**

**Test Summary Report :-**

**Project Name :-** Online Learning Platform

**Testing Period :-** 01-01-0000 To 10-00-0000

**Total Test Cases :-** 10

**Executed Test Cases:-** 10

**Passed Test Cases:-** 9

**Failed Test Cases:-** 1

**Blocked Test Cases:-** 0

**Defects Identified**

Major Defects: 1

Minor Defects: 2

Critical Defects: 0

**Over all Test Status :-**

Most of the application features are working correctly. The identified defects have been reported to the development team.

**Release Recommendation :-**

The application is recommended for release after fixing the reported Major defect and completing regression testing.